Undergraduate studies

Applied Mathematics with Scientific Computing and Scientific Machine Learning (Bachelor of Mathematics – Honours)

Systems of Study

Co-operative
Regular

Declaration Requirements

Before declaring this academic plan, see invalid credential combinations.

Minimum Average(s) Required

- A minimum cumulative overall average of 60.0%.
- A minimum cumulative major average of 65.0%: all math courses.

Graduation Requirements

- See Bachelor of Mathematics degree-level requirements.
- Complete all the required courses listed below.
- Complete a minimum of 13.0 units of math courses.
- Complete a minimum of 5.0 units of non-math courses.

Co-operative Education Program Requirements

For students in the co-operative system of study, see Bachelor of Mathematics co-operative education program requirements.

Course Requirements

Required Courses

- Complete all of the following
  - Complete all the following:
    - AMATH231 - Calculus 4 (0.50)
    - AMATH445 - Scientific Machine Learning (0.50)
    - CS234 - Data Types and Structures (0.50)
  - Complete 1 of the following:
    - AMATH242 - Introduction to Computational Mathematics (0.50)
    - CS371 - Introduction to Computational Mathematics (0.50)
  - Complete 1 of the following:
    - AMATH250 - Introduction to Differential Equations (0.50)
    - AMATH251 - Introduction to Differential Equations (Advanced Level) (0.50)
  - Complete 1 of the following:
    - AMATH342 - Computational Methods for Differential Equations (0.50)
    - AMATH345 - Data-Driven Mathematical Models (0.50)
    - AMATH449 - Neural Networks (0.50)
    - CS479 - Neural Networks (0.50)
  - Complete 1 of the following:
    - CO250 - Introduction to Optimization (0.50)
    - CO255 - Introduction to Optimization (Advanced Level) (0.50)
  - Complete 1 of the following:
    - MATH237 - Calculus 3 for Honours Mathematics (0.50)
    - MATH247 - Calculus 3 (Advanced Level) (0.50)
  - Complete 2 additional AMATH courses at the 300- or 400-level
  - Complete 1 additional AMATH course at the 400-level
  - Complete 4 additional courses from the options in List 1

List 1

- Complete 4 of the following:
  - AMATH342 - Computational Methods for Differential Equations (0.50)
  - AMATH345 - Data-Driven Mathematical Models (0.50)
  - AMATH391 - Data Analysis with Fourier and Wavelet Methods (0.50)
  - AMATH442 - Computational Methods for Partial Differential Equations (0.50)
  - AMATH446 - Introduction to Mathematics of Deep Learning (0.50)
  - AMATH449 - Neural Networks (0.50)
  - AMATH477 - Stochastic Processes for Applied Mathematics (0.50)
  - CO367 - Nonlinear Optimization (0.50)
  - CO466 - Continuous Optimization (0.50)
  - CO481 - Introduction to Quantum Information Processing (0.50)
  - CS231 - Algorithmic Problem Solving (0.50)
  - CS467 - Introduction to Quantum Information Processing (0.50)
  - CS475 - Computational Linear Algebra (0.50)
  - CS479 - Neural Networks (0.50)
  - PHYS467 - Introduction to Quantum Information Processing (0.50)
  - PMATH343 - Introduction to the Mathematics of Quantum Information (0.50)
  - STAT331 - Applied Linear Models (0.50)
  - STAT341 - Computational Statistics and Data Analysis (0.50)
  - STAT441 - Statistical Learning - Classification (0.50)
  - STAT444 - Statistical Learning - Advanced Regression (0.50)

Additional Constraints

1. Courses used to satisfy core program requirements cannot be used to satisfy requirements on List 1.
2. Students may only complete one course from any cross-listed set.

Offered by Faculty(ies)

Faculty of Mathematics

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The University of Waterloo acknowledges that

much of our work takes place on the traditional territory of the Neutral, Anishinaabeg, and Haudenosaunee peoples. Our main campus is situated on the Haldimand Tract, the land granted to the Six Nations that includes six miles on each side of the Grand River. Our active work toward reconciliation takes place across our campuses through research, learning, teaching, and community building, and is co-ordinated within the Office of Indigenous Relations <https://uwaterloo.ca/indigenous>.



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